

AERO40009 Thermodynamics

Tutorial Sheet 3

REVERSIBILITY, SECOND LAW, CLAUSIUS INEQUALITY

1. Indicate whether the following processes are reversible or irreversible. Neglect heat transfer unless the contrary is indicated. In one irreversible case devise a perpetual motion machine of the second kind (i.e. one which violates the basic second law statement) which would result if the process were reversible.
- IR** (a) The temperature of a mass of gas confined in a rigid container is raised by rotating a fan impeller inside the mass of gas.
- IR** (b) Some water confined under a piston of constant weight is stirred until half of it evaporates.
- R** (c) Some water confined as in (b) is evaporated by a slow addition of heat.
- IR** (d) A heavy piston confining a gas in vertical cylinder is held in position by stops. When the stops are removed the piston falls because it has much more weight than the initial pressure of the gas can support.
- IR** (f) The fluid stream leaving the fan in (e) is introduced into a large room where it eddies around until it leaves the room slowly through a large opening.
- IR** (g) A stream of water is raised in temperature from 10°C to 90°C by mixing it with a stream of steam at 100°C.
- R** (h) A stream of fluid passes through ideally shaped fan blades without frictional resistance and without pressure drop. An observer standing on the fan blade sees no change in the stream speed between entrance and exit. A stationary observer sees a marked increase in stream speed between entrance and exit

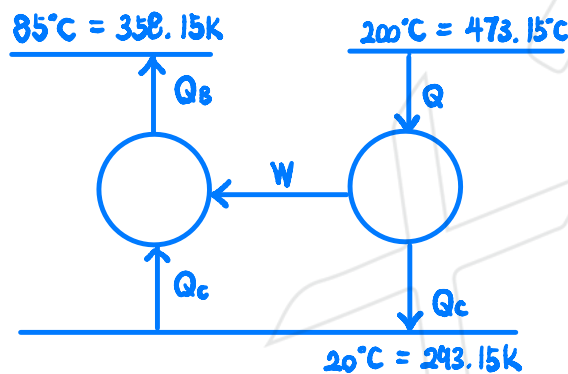
[Explanation for velocity increase done in tutorials.](#)
[See next year velocity triangles](#)

2. In order to heat a building, heat must be supplied at 85°C . Q units of heat are freely available at 200°C as waste heat from a near-by processing plant. The surroundings are at 20°C . If reversible devices were available, how much heat could be transferred to the building?

Ans. $Q_B = 2.09 Q$

$$\eta = 1 - \frac{T_c}{T_H} = 1 - \frac{Q_B}{Q}$$

$$\therefore Q_B = Q \frac{T_c}{T_H} = Q \frac{85^{\circ}\text{C}}{200^{\circ}\text{C}} = Q \frac{358.15\text{K}}{473.15\text{K}} = 0.757 Q$$



$$W = Q\eta = Q \left(1 - \frac{T_c}{T_H} \right) = 0.380 Q$$

$$Q_B = W + Q_c$$

$$\therefore \frac{Q_B}{T_b} = \frac{Q_c}{T_c}$$

$$\therefore Q_B = W + \frac{T_c}{T_b} Q_B$$

$$\therefore Q_B = \frac{W}{\left(1 - \frac{T_c}{T_b} \right)} = 2.096 Q$$

3. (a) One kg of water at 273 K is brought into contact with a heat reservoir at 373 K. When the water has reached 373K, what are the entropy changes of the water, the reservoir and the universe?
- (b) If the water had been heated by first bringing it into contact with a reservoir at 323K until the water reached 323K and then with a reservoir at 373K until it reached 373K, what would be the change in entropy of the universe?
- (c) How could the water be heated from 273K to 373K with no change in the entropy of the universe?

Ans. (a) +1.308 kJ/K, -1.123 kJ/K, +0.185 kJ/K; (b) +0.098 kJ/K.

Take the specific heat of water as 4.19 kJ/kg K.

$$C_{v, \text{water}} = 4.19 \text{ kJ/kg} \cdot \text{K}$$

$$\begin{aligned} (a) \Delta S_{\text{water}} &\cong \int \frac{dQ}{T} = MC_v \ln\left(\frac{T_i}{T_o}\right) + MR \ln\left(\frac{V_i}{V_o}\right) \\ &= MC_v \ln\left(\frac{T_i}{T_o}\right) \\ &= 1.3077 \text{ kJ/K} \end{aligned}$$

$$\Delta S_{\text{surroundings}} = -\frac{dQ}{T_o} = -\frac{MC_v \Delta T}{T_o} = -1.1233 \text{ kJ/K}$$

$$\Delta S_{\text{universe}} = 1.3077 \text{ kJ/K} - 1.1233 \text{ kJ/K} = 0.1844 \text{ kJ/K}$$

$$\begin{aligned} (b) \Delta S_{\text{universe}}' &= MC_v \left[\ln\left(\frac{T_{1'}}{T_o}\right) + \ln\left(\frac{T_2}{T_{1'}}\right) \right] - \frac{MC_v (T_{1'} - T_o)}{T_{1'}} - \frac{MC_v (T_2 - T_{1'})}{T_2} \\ &= 1.3077 \text{ kJ/K} - 0.6486 \text{ kJ/K} - 0.5617 \text{ kJ/K} \\ &= 0.0974 \text{ kJ/K} \end{aligned}$$

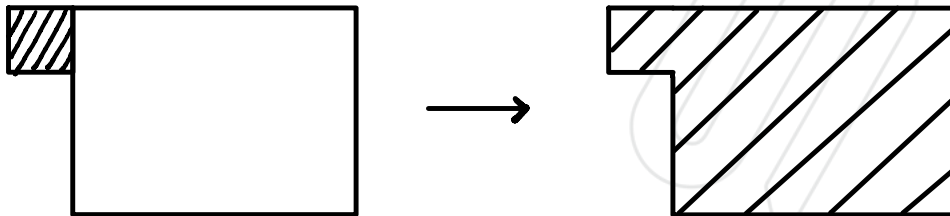
(c) To bring water into contact with infinite reservoir, where there is infinitesimal increase in temperature between reservoirs.

4. For a system consisting of a pure substance, indicate whether the following expressions apply to all reversible processes or cycles, to all irreversible ones, or to both reversible ones and irreversible ones.
(No appreciable velocities are involved).

	reversible	irreversible	both
(a) $dS = \frac{dQ}{T}$	✓		
(b) $dQ = du + pdv$	✓		
(c) $\eta = \frac{T_1 - T_2}{T_1}$ $T_2 = T_C; T_1 = T_H$	✓		
(d) $\eta = \frac{W}{Q_{in}}$			✓
(e) $dh = du + d(pv)$			✓
(f) $Q = \Delta E + W$			✓
(g) $dS > \int \frac{dQ}{T}$		✓	
(h) $TdS = du + pdv$	✓		
(i) $\oint dS = 0$ <i>entropy is a property!</i>	✓		✓
(j) $\frac{dQ}{T} < ds$		✓	

5. A rigid insulated container is divided by a partition into parts of volume 1 m^3 and 9 m^3 respectively, with the smaller part containing air at 2.87 bar and 300 K and the larger part being empty. The partition is then removed, allowing the air to occupy the entire volume.

Calculate the increase in the entropy of the air, considered as a perfect gas directly from the Boltzmann equation.



$$\therefore V \rightarrow 10V \quad \therefore \Omega \rightarrow 10^N \Omega$$

No change in momentum

$$\therefore \Delta S = k_B \ln(10^N \Omega) - k_B \ln \Omega$$

$$= k_B \ln(10^N)$$

$$= N k_B \ln 10$$

$$= MR \ln 10$$

$$= \frac{P_1 V_1}{T_1} \ln 10$$

$$= 2202.81 \text{ J/K}$$

6. A rigid insulated container is divided into 2 halves of equal volume 1m^3 by a diaphragm. One half contains air at 10 bar, 500K the other contains air at 3 bar, 300K. The diaphragm is suddenly ruptured.

- Calculate the final temperature and pressure. and the increase in entropy of the air.
- Calculate the increase in entropy of the universe after the diaphragm is ruptured.

Ans: (a): $T_F = 433.3\text{K}$, $P_F = 6.5\text{ bar}$

- (b): Hints: denote by subscript "1" the half initially at 10 bar, "2" the other half and let "3" represent the common final state. Then:

$$m_1 = 6.969\text{kg}, m_2 = \dots \Rightarrow \Delta S_{1F} = -144.4\text{J/K}, \\ \Delta S_{2F} = +512.4\text{J/K}; \Delta S_{\text{universe}} = +368\text{ J/K}$$



$$(a) P_F = \frac{1}{2} (P_1 + P_2) = 6.5\text{ bar}$$

$$\therefore PV = MRT \quad \therefore M_1 R = \frac{P_1 V_1}{T_1} = 0.02, M_2 R = 0.01$$

$$\therefore MR = (M_1 + M_2) R = 0.03 \quad \therefore T_F = \frac{P_F V}{MR} = 433.3\text{K}$$

$$\Delta S = MC_v \ln \left(\frac{T_1 T_2}{T_0^2} \right) + MR \ln \left(\frac{V_1 V_2}{V_0^2} \right)$$

=

6. A rigid insulated container is divided into 2 halves of equal volume 1m^3 by a diaphragm. One half contains air at 10 bar, 500K the other contains air at 3 bar, 300K. The diaphragm is suddenly ruptured.

- (a) Calculate the final temperature and pressure. and the increase in entropy of the air.
 (b) Calculate the increase in entropy of the universe after the diaphragm is ruptured.

Ans: (a): $T_F = 433.3\text{K}$, $P_F = 6.5\text{ bar}$

(b): Hints: denote by subscript "1" the half initially at 10 bar, "2" the other half and let "3" represent the common final state. Then:

$$m_1 = 6.969\text{kg}, m_2 = \dots \Rightarrow \Delta S_{1F} = -144.4\text{J/K},$$

$$\Delta S_{2F} = +512.4\text{J/K}; \Delta S_{\text{universe}} = +368\text{ J/K}$$

$$= \Delta S_{\text{air}} = \Delta S_u \text{ as it is insulated}$$

$$(Q) \quad m_1 C_V T_1 + m_2 C_V T_2 = (m_1 + m_2) C_V T_F$$

$$\therefore \frac{P_1 V_1}{RT_1} C_V T_1 + \frac{P_2 V_2}{RT_2} C_V T_2 = \left(\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2} \right) C_V T_F$$

$$\therefore T_F = \frac{\left(\frac{P_1 V_1}{R} + \frac{P_2 V_2}{R} \right)}{\left(\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2} \right)} = \frac{P_1 V_1 + P_2 V_2}{\left(\frac{P_1 V_1}{T_1} + \frac{P_2 V_2}{T_2} \right)} = 433.3\text{K}$$

$$\therefore P_F = \frac{(m_1 + m_2) R T_F}{V_F} = 6.5\text{ bar}$$

$$\Delta S_{1 \rightarrow 3} = m_1 \left[C_V \ln \left(\frac{T_F}{T_1} \right) + R \ln \left(\frac{V_F}{V_1} \right) \right] = -144.4\text{ J/K}$$

$$\Delta S_{2 \rightarrow 3} = m_2 \left[C_V \ln \left(\frac{T_F}{T_2} \right) + R \ln \left(\frac{V_F}{V_2} \right) \right] = 512.4\text{ J/K}$$

$$\therefore \Delta S_{\text{air}} = \Delta S_{1 \rightarrow 3} + \Delta S_{2 \rightarrow 3} = 368.0\text{ J/K}$$

$$(b) \quad \Delta S_u = \Delta S_{\text{air}} = 368.0\text{ J/K}$$

7. Show that for one-dimensional steady adiabatic motion of a perfect gas along a duct, the energy equation gives:

$$C_p T + \frac{1}{2} v^2 = \text{constant}.$$

The speed of sound, α , in a perfect gas is given by

$$\alpha^2 = \frac{\gamma P}{\rho} = \gamma R T$$

Hence show that the energy equation can be written as

$$T \left\{ 1 + \frac{\gamma - 1}{2} M^2 \right\} = \text{const.}$$

If the motion is also reversible (i.e. frictionless and isentropic therefore), show that the energy equation can be also written as

$$P \left\{ 1 + \frac{\gamma - 1}{2} M^2 \right\}^{\frac{\gamma}{\gamma - 1}} = \text{const.}$$

and

$$\rho \left\{ 1 + \frac{\gamma - 1}{2} M^2 \right\}^{\frac{1}{\gamma - 1}} = \text{const.}$$

For steady flow, $Q - W_x = m (h_{0, \text{out}} - h_{0, \text{in}})$

$$\therefore 0 = m (C_p T_{\text{out}} + \frac{1}{2} v_{\text{out}}^2 - C_p T_{\text{in}} - \frac{1}{2} v_{\text{in}}^2)$$

$$\therefore C_p T_{\text{out}} + \frac{1}{2} v_{\text{out}}^2 = C_p T_{\text{in}} + \frac{1}{2} v_{\text{in}}^2$$

$$\therefore C_p T + \frac{1}{2} v^2 = \text{const.}$$

$$\therefore C_p T + \frac{1}{2} \alpha^2 M^2 = \text{const.}$$

$$\therefore \alpha^2 = \frac{\gamma P}{\rho} = \gamma R T$$

$$\therefore C_p T + \frac{1}{2} \delta R T M^2 = T (C_p + \frac{1}{2} \delta (C_p - C_v) M^2) = \text{const.}$$

$$\therefore T [1 + \frac{1}{2} \delta (1 - \frac{1}{\gamma}) M^2] = T [1 + \frac{\delta - 1}{2} M^2] = \text{const.}$$

If motion is reversible,

$$\therefore \frac{PV}{MR} [1 + \frac{\delta - 1}{2} M^2] = \frac{P}{\rho R} [1 + \frac{\delta - 1}{2} M^2] = \text{const.}$$

$$\therefore \frac{P}{\rho} [1 + \frac{\delta - 1}{2} M^2] = \text{const.}$$

$$\therefore P \gamma^{\frac{1}{\delta}} = \frac{P}{\rho^{\frac{1}{\delta}}} = \text{const.}$$

$$\therefore \rho^{\frac{1}{\delta}} = \frac{P}{\text{const.}} \quad \therefore \rho = \left(\frac{P}{\text{const.}} \right)^{\delta}$$

$$\therefore \frac{P}{\left(\frac{P}{\text{const.}} \right)^{\delta}} [1 + \frac{\delta - 1}{2} M^2] = \text{const.}$$

$$\therefore P^{\frac{\delta - 1}{\delta}} [1 + \frac{\delta - 1}{2} M^2] = \text{const.}$$

$$\therefore P [1 + \frac{\delta - 1}{2} M^2]^{\frac{\delta}{\delta - 1}} = \text{const.}$$

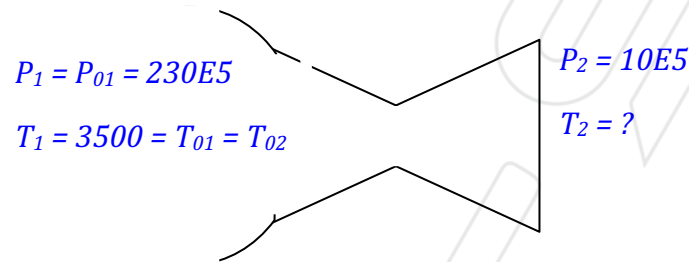
$$\therefore \rho^{\delta} = \frac{P}{\text{const.}}$$

$$\therefore \rho^{\delta} [1 + \frac{\delta - 1}{2} M^2]^{\frac{\delta}{\delta - 1}} = \text{const.}$$

$$\therefore \rho [1 + \frac{\delta - 1}{2} M^2]^{\frac{1}{\delta - 1}} = \text{const.}$$

8. In the combustion chamber of a rocket motor the gases are at a temperature of 3500 K and a pressure of 230 bar. The velocity can be regarded as negligible. The gases expand through a nozzle to an exit pressure of 10 bar. Assuming steady, adiabatic frictionless motion calculate the exit Mach number, velocity, density and temperature.

Assume perfect gas behaviour with $\gamma = 1.23$, $R = 360 \text{ J/kg K}$.



(Ans: 2.63, 1947K, 2442 m/s, 1.43 kg/m³).

$$\therefore P \left[1 + \frac{\gamma-1}{2} M^2 \right]^{\frac{\gamma}{\gamma-1}} = \text{const.}$$

$$\therefore P_1 \left[1 + \frac{\gamma-1}{2} M_1^2 \right]^{\frac{\gamma}{\gamma-1}} = P_2 \left[1 + \frac{\gamma-1}{2} M_2^2 \right]^{\frac{\gamma}{\gamma-1}}$$

$$\therefore M_1 = 0 \quad \therefore M_2 = 2.633$$

$$\therefore T \left[1 + \frac{\gamma-1}{2} M^2 \right] = \text{const.}$$

$$\therefore T_2 = \frac{T_1}{\left[1 + \frac{\gamma-1}{2} M_2^2 \right]} = 1947.410 \text{ K}$$

$$\therefore a^2 = \frac{\gamma P}{\rho} = \gamma R T$$

$$\therefore V_2 = a M = \sqrt{\gamma R T} M = 2445.025 \text{ m/s}$$

$$\therefore \rho_2 = \frac{P_2}{R T_2} = 1.426 \text{ kg/m}^3$$